
Tangent Schrödinger Bridge Matching: Learning Stochastic Transport with Mechanistic Sensitivities

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Abstract

1 Diffusion Schrödinger bridges learn stochastic transports between endpoint distri-
2 butions while remaining close to a reference diffusion, enabling flexible modeling
3 of stochastic transformations between observed states. However, a learned bridge
4 can match observed endpoint distributions while responding incorrectly when
5 a physical condition or parameter is perturbed. We introduce TANGENT-SBM,
6 which propagates intervention sensitivities alongside trajectories of the learned
7 conditional SDE, allowing externally supplied sensitivity information to constrain
8 how the learned stochastic dynamics respond when u changes. When a sensitivity
9 target is defined for each stochastic realization, we match it directly. When only
10 the conditional mean sensitivity is known, we use two independent stochastic
11 trajectories to match that mean without explicitly penalizing legitimate variation
12 across trajectories. Across an analytically controlled Gaussian system, a stochastic
13 double well, and the official PDEBench two-dimensional diffusion–reaction simu-
14 lator extended with parameter interventions, TANGENT-SBM consistently reduces
15 intervention-sensitivity error. On PDEBench, it reduces directional sensitivity
16 error by 37–44% relative to Conditional DSBM while maintaining comparable
17 terminal-field accuracy.

18 1 Introduction

19 Schrödinger bridges (SBs) model stochastic transformations between endpoint distributions while
20 remaining close to a reference stochastic process [Léonard, 2014, De Bortoli et al., 2021]. Diffusion-
21 based neural solvers, including Diffusion Schrödinger Bridge Matching (DSBM), make such trans-
22 ports practical for complex high-dimensional distributions [Wang et al., 2021, Shi et al., 2023, Tong
23 et al., 2024]. This makes them useful when one wants to learn not only a mapping between observed
24 states, but a stochastic process connecting them. Related diffusion, score-based, flow-matching,
25 neural-SDE, and stochastic-interpolant models provide other continuous-time approaches to learned
26 transport [Ho et al., 2020, Song et al., 2021, Lipman et al., 2023, Kidger et al., 2021, Albergo et al.,
27 2023].

28 Scientific modeling, however, frequently asks a stronger question: after fitting the observed states, will
29 the learned dynamics respond correctly when a physical parameter, forcing, control, or intervention
30 changes?

31 Let u denote such an external variable. A conditional bridge can be given access to u , allowing its
32 dynamics to vary across operating conditions. Conditioning alone, however, does not identify how
33 the learned transport should change when u is perturbed [Shi et al., 2022]. A model may therefore
34 fit the conditional endpoint distribution accurately at observed parameter settings while learning an
35 incorrect local response to the parameter. Two transports can therefore agree on endpoint observations

yet imply different local intervention responses and different extrapolations to unobserved parameter settings. This matters in scientific settings where u may represent diffusivity, a reaction rate, source strength, or forcing: a useful model should predict not only the state, but also whether and by how much that state changes when the physical parameter changes.

The issue is that intervention response need not be identifiable from endpoint values alone. Endpoint observations constrain function values at sampled operating points, whereas scientific knowledge may additionally constrain derivatives of the induced stochastic transport. Our conditional-Gaussian benchmark makes this gap explicit: the endpoint training anchors make an entire sinusoidal component invisible in value space while leaving its derivative nonzero. The same issue appears empirically in more complex systems: Conditional DSBM can achieve reasonable endpoint accuracy while learning sensitivities that are less accurate than the trivial assumption that the output does not change with u . Thus, *conditioning specifies what the bridge may depend on; it does not determine how the bridge should respond when that condition changes.*

We address this ambiguity with TANGENT-SBM. Rather than constraining only the states reached by a conditional stochastic bridge, TANGENT-SBM also tracks how the learned trajectory changes when the external condition is perturbed. Externally supplied mechanistic sensitivities can therefore supervise the response of the learned stochastic dynamics themselves. This provides information that endpoint observations alone may not identify.

A second issue arises when the scientific target is an *expected* response rather than a pathwise one. For a stochastic bistable system, individual tangent trajectories legitimately vary with the Brownian realization. Matching every realization to a mean target with ordinary MSE therefore suppresses stochastic response variability. We derive a two-independent-rollout estimator that isolates mean-response error and empirically show that it retains substantially more simulator tangent variance than naive single-rollout MSE.

Our main contributions are:

- a tangent-lift formulation for turning externally supplied mechanistic sensitivities into supervision for a conditional stochastic transport;
- sensitivity-supervision objectives for two forms of scientific information: realization-level sensitivities when they are available, and conditional-mean sensitivities when the response itself is stochastic, using a double-sampling construction to avoid explicitly penalizing response variance;
- controlled evidence that endpoint conditioning need not identify intervention response, from an analytic construction and a stochastic bistable system to high-dimensional PDEBench dynamics;
- controlled experiments isolating sensitivity-target correctness, supervision amount, extrapolation, directionality, finite-difference response matching, and simulator-budget allocation.

2 Tangent Schrödinger Bridge Matching

Consider the conditional learned SDE

$$dX_t = b_\theta(X_t, u, t) dt + \sigma_t dW_t. \quad (1)$$

Here $u \in \mathbb{R}^d$ is the intervention variable. To measure how the state changes when u is perturbed in a direction $v \in \mathbb{R}^d$, define the directional sensitivity

$$R_t^v = \frac{\partial X_t}{\partial u} v.$$

Assuming additive noise whose scale σ_t does not depend on X_t or u , differentiating Eq. (1) with respect to u gives

$$dR_t^v = [\nabla_X b_\theta(X_t, u, t) R_t^v + \nabla_u b_\theta(X_t, u, t) v] dt. \quad (2)$$

If X_0 is independent of u , as in our experiments, then $R_0^v = 0$. The resulting tangent lift evolves the state and its directional sensitivity jointly:

$$d \begin{bmatrix} X_t \\ R_t^v \end{bmatrix} = \begin{bmatrix} b_\theta \\ \nabla_X b_\theta R_t^v + \nabla_u b_\theta v \end{bmatrix} dt + \begin{bmatrix} \sigma_t I \\ 0 \end{bmatrix} dW_t. \quad (3)$$

Sensitivity information can therefore modify the learned drift through the stochastic trajectory rather than acting only as a derivative penalty on a terminal predictor.

Tangent-SBM overview

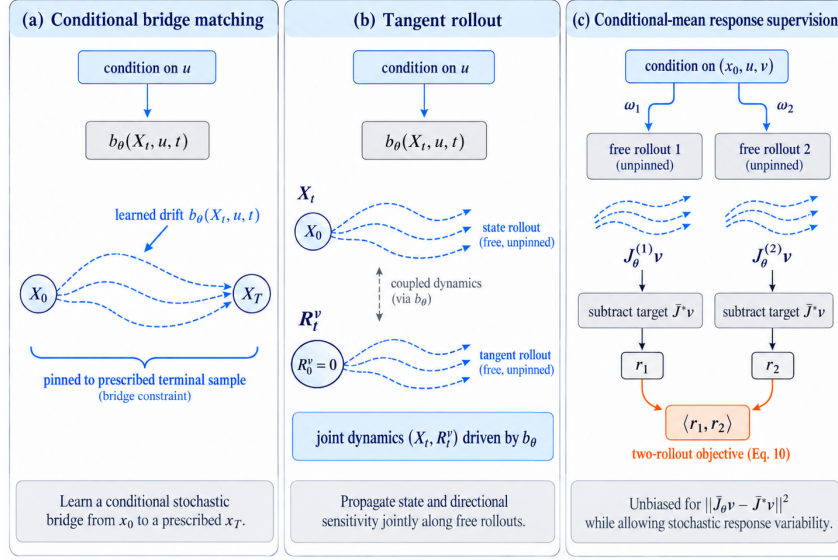


Figure 1: **Tangent-SBM overview.** (a) Conditional bridge matching uses endpoint-pinned trajectories. (b) TANGENT-SBM jointly propagates state and directional sensitivity on free rollouts; pathwise targets are matched directly. (c) Conditional-mean targets use two independent rollouts and the residual cross-term objective.

80 **Sensitivity of a terminal quantity.** The final state X_T need not itself be the only quantity of interest.
 81 Let

$$Y_\theta = h(X_T, u)$$

82 denote a differentiable terminal quantity derived from the learned trajectory. For a particular Brownian
 83 realization ω , its sensitivity to the intervention is

$$J_\theta(u, \omega) = \frac{\partial Y_\theta(u, \omega)}{\partial u}. \quad (4)$$

84 At terminal time, the sensitivity of $Y_\theta = h(X_T, u)$ in direction v follows from the chain rule:

$$J_\theta v = \nabla_X h(X_T, u) R_T^v + \nabla_u h(X_T, u) v. \quad (5)$$

85 The first term captures the effect of u through the terminal state X_T ; the second captures any direct
 86 dependence of h on u .

87 In high dimensions we do not construct the full Jacobian J_θ . Instead we compute the product $J_\theta v$
 88 directly using a Jacobian–vector product (JVP), i.e., the directional derivative of the output when u is
 89 perturbed along v [Pearlmutter, 1994].

90 **Realization-level (pathwise) sensitivity supervision.** In some systems the desired sensitivity is well
 91 defined for each individual realization. For example, the reference simulator may be deterministic
 92 once its initial condition and physical parameters are fixed, or the sensitivity may be independent of
 93 the stochastic perturbation. In this case, the model sensitivity for each trajectory can be compared
 94 directly with a target $J^*(u, \omega)$:

$$\Phi_{\text{path}} = \mathbb{E}_\omega \left[\|J_\theta(u, \omega) - J^*(u, \omega)\|_F^2 \right]. \quad (6)$$

95 This objective asks each learned trajectory to have the appropriate local response to u .

96 **Conditional-mean sensitivity supervision.** In other stochastic systems, the sensitivity itself varies
 97 across random realizations. The double-well benchmark is an example: even with the same initial

state x_0 and intervention u , different Brownian paths produce different terminal sensitivities. In that setting the scientific target may be the average response

$$\bar{J}_\theta(x_0, u) = \mathbb{E}_\omega [J_\theta(u, \omega) \mid X_0 = x_0, u], \quad (7)$$

rather than the sensitivity of any particular trajectory. We obtain the corresponding simulator target

$$\bar{J}^*(x_0, u) \approx \mathbb{E}_\omega [J_{\text{sim}}(x_0, u, \omega) \mid X_0 = x_0, u]. \quad (8)$$

A naive approach would compare every random model trajectory with the same mean target using squared error. However,

$$\mathbb{E}_\omega \|J_\theta - \bar{J}^*\|_F^2 = \|\bar{J}_\theta - \bar{J}^*\|_F^2 + \mathbb{E}_\omega \|J_\theta - \bar{J}_\theta\|_F^2. \quad (9)$$

The first term is the mean-sensitivity error that we want to minimize, whereas the second penalizes variability across otherwise valid stochastic trajectories.

To remove this explicit variance penalty, we generate two independent trajectories using the same (x_0, u, v) but independent Brownian noise, and use

$$\hat{\Phi}_{\text{mean}} = \left\langle J_\theta^{(1)} - \bar{J}^*, J_\theta^{(2)} - \bar{J}^* \right\rangle_F. \quad (10)$$

Because the two trajectories are conditionally independent,

$$\mathbb{E}[\hat{\Phi}_{\text{mean}} \mid x_0, u, v] = \|\bar{J}_\theta - \bar{J}^*\|_F^2.$$

This is the classical double-sampling idea [Baird, 1995]; here we use it to supervise the conditional mean sensitivity of a learned stochastic transport. The estimator removes the explicit variance penalty above, but does not directly force the model variance to match the simulator variance. Although individual cross terms can be negative, training uses their minibatch average over 256 independent rollout pairs per response update; all three reported double-well runs remained numerically stable.

Training objective and unpinned response rollouts. We optimize

$$\mathcal{L}_{\text{Tangent}} = \mathcal{L}_{\text{BM}} + \lambda_{\text{sens}} \Phi_{\text{sens}}. \quad (11)$$

The distinction between the two terms is operationally important. $\mathcal{L}_{\text{BM}}$ is computed from the bridge-matching construction used by DSBM, whereas Φ_{sens} is evaluated on free trajectories of the current learned SDE without conditioning on a prescribed terminal sample; we refer to these as *unpinned* rollouts. Consequently, the tangent term measures the response of the model that will actually be simulated at test time, not the derivative of an endpoint-pinned reciprocal interpolation. During a response update we sample (x_0, u, v) , roll out the learned SDE, propagate R_t^v alongside X_t , compare $J_\theta v$ with the mechanistic target, and differentiate through the JVP trajectory into the drift parameters. For conditional-mean sensitivity targets, the same (x_0, u, v) seeds two independent Brownian rollouts in Eq. (10).

Directional sensitivity supervision. For $u \in \mathbb{R}^d$ and an m -dimensional terminal quantity, the sensitivity is an $m \times d$ Jacobian. Rather than constructing it explicitly, we sample $v \in \mathbb{R}^d$ and compare $J_\theta v$ with $J^* v$. Writing $A = J_\theta - J^*$,

$$\mathbb{E} \|Av\|_2^2 = \|A\|_F^2 \quad \text{for } v \sim \mathcal{N}(0, I_d), \quad \mathbb{E} \|Av\|_2^2 = \|A\|_F^2 / d \quad \text{for unit-sphere } v.$$

Thus random directions provide a matrix-free probe of Jacobian error. For PDEBench, $d = 3$ while the output has $2 \times 128 \times 128 = 32768$ values, so we sample unit intervention directions and report directional sensitivity error directly.

3 Experimental Setup

Conditional Gaussian. We use

$$X_T = AX_0 + B_1 u + B_2 u^2 + B_3 \sin(\pi u) + \epsilon, \quad (12)$$

for scalar u and Gaussian ϵ , with exact response $J_f^*(u) = B_1 + 2B_2 u + \pi B_3 \cos(\pi u)$. Endpoint training uses only $u \in \{-1, 0, 1\}$, where the sinusoidal value vanishes although its derivative does not. This construction is deliberately designed as an existence proof that endpoint values need not identify intervention sensitivity; it is not evidence that this pathology is prevalent in every conditional transport problem. The evaluation regions are defined together with the other benchmarks below.

136 **Stochastic double well.** We simulate

$$dX_t = [X_t - X_t^3 + 0.30 \tanh(u)]dt + 0.45 dW_t, \quad (13)$$

137 whose tangent obeys $dR_t/dt = (1 - 3X_t^2)R_t + 0.30 \operatorname{sech}^2(u)$. Because the Brownian forcing
138 changes the state trajectory, it also changes the terminal sensitivity R_T . Thus there is no single
139 realization-level sensitivity associated with a given (x_0, u) . We instead use the conditional mean
140 sensitivity

$$\mathbb{E}[R_T \mid x_0, u].$$

141 For each (x_0, u) , this target is estimated by repeatedly simulating the double-well system with
142 independent Brownian noise and averaging the resulting terminal sensitivities. Training then uses the
143 conditional-mean objective in Eq. (10).

144 **PDEBench diffusion–reaction.** Our high-dimensional benchmark uses the official two-dimensional
145 diffusion–reaction simulator from PDEBench [Takamoto et al., 2022]. The state consists of two
146 interacting spatial fields, $q_1(x, y, t)$ and $q_2(x, y, t)$, defined on a 128×128 grid. Their evolution is

$$\partial_t q_1 = q_1 - q_1^3 - k - q_2 + D_u \nabla^2 q_1, \quad (14)$$

$$\partial_t q_2 = q_1 - q_2 + D_v \nabla^2 q_2. \quad (15)$$

147 The local reaction terms determine how the two fields interact at each spatial location, while the
148 Laplacian terms $\nabla^2 q_1$ and $\nabla^2 q_2$ spread each field spatially. The coefficients D_u and D_v control the
149 two diffusion rates, and k is a reaction parameter.

150 We treat these three physical coefficients as interventions. Rather than perturbing their raw values
151 directly, we introduce $a = (a_{D_u}, a_{D_v}, a_k) \in \mathbb{R}^3$ and set

$$D_u = D_u^0 2^{a_{D_u}}, \quad D_v = D_v^0 2^{a_{D_v}}, \quad k = k^0 2^{a_k}.$$

152 Thus $a_j \in \{-1, 0, 1\}$ corresponds to halving, retaining, or doubling the associated coefficient;
153 endpoint training uses all $3^3 = 27$ combinations.

154 Let $F(X_0, a)$ denote the terminal field produced by the PDE simulator from initial field X_0 at
155 parameter setting a . To obtain a sensitivity target, we choose a random unit direction $r \in \mathbb{R}^3$ and
156 approximate the directional derivative by the centered finite difference

$$J^* r \approx \frac{F(X_0, a + \epsilon r) - F(X_0, a - \epsilon r)}{2\epsilon}, \quad \epsilon = 0.03. \quad (16)$$

157 Both calls share the same simulator seed so that their difference isolates the parameter perturbation.
158 We separately evaluate whether the learned local sensitivity predicts the effect of a larger finite
159 perturbation $\Delta a = 0.25$. Our PDEBench experiment is intended to evaluate intervention-response
160 fidelity within a learned stochastic transport, rather than to compete with deterministic operator-
161 learning models trained directly as parameter-to-solution regressors; the objectives and data regimes
162 are therefore not directly comparable.

163 **Baselines and training protocol.** We compare three main models. DSBM does not receive the
164 intervention variable and therefore provides a zero-sensitivity reference. CONDITIONAL DSBM
165 receives the intervention u (or a for PDEBench) but is trained only with the standard bridge-matching
166 objective. TANGENT-SBM receives the same intervention and additionally uses the sensitivity targets
167 described above. The key comparison is therefore Conditional DSBM versus TANGENT-SBM: both
168 know the physical parameter, but only TANGENT-SBM is told how the output should change when
169 that parameter is perturbed.

170 Within each benchmark, Conditional DSBM and TANGENT-SBM start from the same conditional
171 checkpoint and use the same continuation training schedule, so their comparison does not depend on
172 different initial models or different numbers of standard continuation updates. Because the bridge
173 and sensitivity losses have different scales across benchmarks, we tune λ_{sens} separately for each
174 benchmark and then fix the selected value for all three-seed results. The sweep ranges and selected
175 values are reported in Appendix D.

176 For PDEBench, constructing sensitivity targets requires 648 additional simulator calls. To test whether
177 the gain is simply due to this extra simulator information, we give Conditional DSBM the same
178 budget as ordinary endpoint data. COND.+LOCALBUDGET uses the exact $\pm \epsilon r$ simulator outputs
179 used to form the sensitivity targets, but treats them as endpoint pairs. COND.+SPREADBUDGET
180 instead spends the same 648 calls on endpoint examples distributed broadly over $[-1.3, 1.3]^3$. We
181 report two training lengths for the spread variant; details are given in Appendix B.

Table 1: **Main results.** Mean $\pm$ standard deviation over three seeds. Lower is better. Endpoint metrics are benchmark-specific as described in Sec. 3.

Benchmark / split	Method	Endpoint	E_J	Finite response
Gaussian / ID	DSBM	0.714 ± 0.002	1.000 ± 0.000	0.321 ± 0.000
	Conditional	0.440 ± 0.013	0.866 ± 0.003	0.275 ± 0.001
	TANGENT-SBM	0.249 ± 0.006	0.050 ± 0.005	0.016 ± 0.001
Gaussian / Far	DSBM	1.397 ± 0.006	1.000 ± 0.000	0.512 ± 0.000
	Conditional	0.397 ± 0.028	0.664 ± 0.008	0.329 ± 0.004
	TANGENT-SBM	0.260 ± 0.009	0.137 ± 0.026	0.085 ± 0.006
Double well / ID	DSBM	0.336 ± 0.005	1.000 ± 0.000	0.110 ± 0.000
	Conditional	0.186 ± 0.006	0.341 ± 0.030	0.035 ± 0.005
	TANGENT-SBM	0.177 ± 0.002	0.250 ± 0.022	0.025 ± 0.003
Double well / Far	DSBM	0.542 ± 0.005	1.000 ± 0.000	0.023 ± 0.000
	Conditional	0.167 ± 0.007	2.549 ± 0.299	0.055 ± 0.006
	TANGENT-SBM	0.166 ± 0.006	0.984 ± 0.216	0.022 ± 0.007
PDEBench / ID	DSBM	0.612 ± 0.034	1.000 ± 0.000	1.000 ± 0.000
	Conditional	0.605 ± 0.038	0.847 ± 0.054	0.843 ± 0.054
	TANGENT-SBM	0.598 ± 0.069	0.508 ± 0.046	0.510 ± 0.046
PDEBench / Far	DSBM	0.853 ± 0.032	1.000 ± 0.000	1.000 ± 0.000
	Conditional	0.729 ± 0.024	0.864 ± 0.065	0.867 ± 0.070
	TANGENT-SBM	0.679 ± 0.066	0.547 ± 0.044	0.548 ± 0.045

Finite-difference response control. On PDEBench, we additionally compare against centered finite-difference response matching under a matched stochastic-rollout budget; implementation details and a perturbation-scale sweep are given in Appendix G.

Sensitivity-supervision regions and evaluation splits. Endpoint observations provide full (X_0, u, X_T) pairs, whereas sensitivity-supervision settings provide only the desired local response to changing u and do not add the corresponding endpoint trajectory to training.

We evaluate four regimes. *Seen* uses fresh samples at endpoint-training parameter values. *ID* uses new values within the range spanned by those anchors. *Near OOD* lies outside the endpoint-training range but inside the sensitivity-supervision region. *Far OOD* lies outside both and therefore tests sensitivity extrapolation. Exact ranges for all benchmarks are reported in Appendix D.

Metrics. We evaluate both terminal-state accuracy and intervention-sensitivity accuracy. For Gaussian and double well, endpoint accuracy is conditional-mean RMSE. For PDEBench, whose output is a spatial field, we use relative ℓ_2 field error.

For a perturbation direction v , the normalized directional sensitivity error is

$$E_J = \frac{\|J_\theta v - J^* v\|_2}{\|J^* v\|_2}. \quad (17)$$

Lower is better. A model that predicts no response to the intervention has $J_\theta v = 0$ and therefore $E_J = 1$. Values below one are better than this zero-sensitivity reference, while values above one are worse.

Because E_J is normalized by the magnitude of the true sensitivity $\|J^* v\|_2$, the same absolute prediction error can produce different values of E_J when the true response is small or large. We therefore compare methods primarily within the same evaluation split.

Finally, we test whether improved local sensitivities translate to finite parameter changes. Gaussian and double well report finite-response RMSE using $\Delta u = 0.25$. PDEBench reports relative ℓ_2 error in the field change induced by $\Delta a = 0.25$. These tests check consistency under a non-infinitesimal perturbation; they are not intended as guarantees for arbitrarily large interventions.

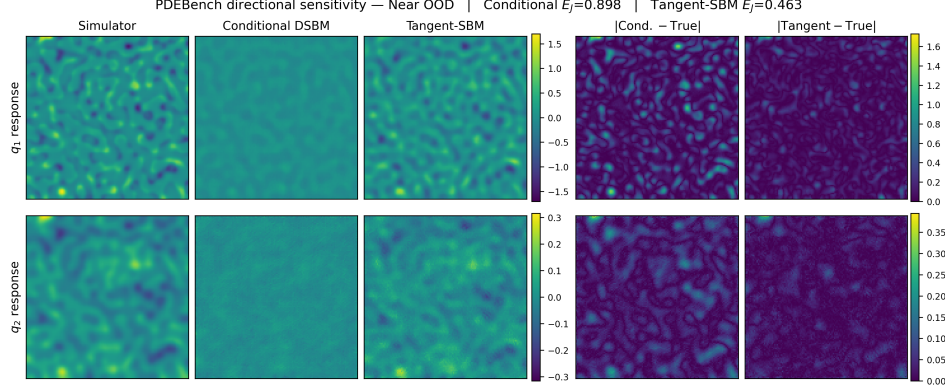


Figure 2: **PDEBench directional sensitivity fields.** Simulator response and predictions for a held-out Near-OOD example selected closest to the median Conditional DSBM E_J over the full Near-OOD evaluation split. TANGENT-SBM better recovers the spatial response structure ($E_J = 0.463$) than Conditional DSBM ($E_J = 0.898$).

4 Results

Endpoint fit does not identify response. The controlled Gaussian experiment isolates the central failure mode. Conditional DSBM improves over unconditional DSBM in endpoint space, yet its ID response error remains $E_J = 0.866$. Tangent-SBM lowers this to 0.050 and simultaneously improves conditional-mean endpoint prediction from 0.440 to 0.249 on ID. The effect persists outside the response-training support: Far-OOD endpoint error improves from 0.397 to 0.260 and E_J from 0.664 to 0.137. Because the sinusoidal term is exactly zero at all endpoint-training anchors, this gap cannot be repaired by fitting those anchor values more accurately; the derivative target supplies information that the endpoint values do not identify.

The double-well experiment shows a different failure mode. Conditional DSBM has reasonable endpoint RMSE but its learned response deteriorates outside the endpoint support, crossing the zero-response reference on Near OOD and reaching $E_J = 2.55$ on Far OOD. Tangent-SBM reduces these values to 0.30 and 0.98 while preserving endpoint accuracy. The Far-OOD value 0.98 should not be interpreted as accurate response recovery: it is close to the null response. The meaningful result there is that Tangent-SBM prevents the severely misdirected response extrapolation produced by conditioning alone.

High-dimensional PDE response. PDEBench provides a substantially harder test: the state is a $2 \times 128 \times 128$ field, the intervention is three-dimensional, and the target is a field-valued directional derivative. Conditional DSBM is only modestly better than the zero-response reference, with E_J between 0.83 and 0.88 across splits. Tangent-SBM reduces this error by approximately 42%, 40%, 44%, and 37% on Seen, ID, Near OOD, and Far OOD, respectively. The corresponding finite-response errors at $\Delta a = 0.25$ show nearly the same reductions, providing a consistency check that the learned local tangent improves the induced finite perturbation. Terminal-field error remains comparable across all four regimes, while Fig. 2 illustrates the corresponding improvement in the spatial structure of the learned directional response. A finite-difference control achieves similar sensitivity accuracy when its perturbation scale is well chosen, while TANGENT-SBM requires no finite-difference scale parameter; the full δ sweep is reported in Appendix G.

Equal-simulator-budget endpoint controls do not recover the sensitivity gains of TANGENT-SBM; LocalBudget and SpreadBudget remain near the Conditional DSBM response error across PDEBench splits (Appendix B).

Conditional-mean sensitivity estimator. The two-rollout construction is not only algebraically unbiased; its statistical effect is visible in the stochastic double well. Naive single-rollout MSE directly penalizes the conditional variance term in Eq. (9). Using the same checkpoint, response data, sensitivity weight, and continuation schedule, the naive objective retains only 31% of simulator tangent variance on ID and 16% on Near OOD. The two-rollout objective retains 67% on both regimes. It also yields lower mean-response RMSE on every split, with substantial gains on Seen, ID,

and Far OOD. The variance ratio is not expected to equal one: our estimator removes the explicit variance penalty but does not itself supervise variance, which remains determined by the learned bridge dynamics.

Sensitivity supervision is data-efficient and target-dependent. Across all three benchmarks, reducing continuous sensitivity supervision to 25% changes response accuracy only modestly. In contrast, shuffled or sign-reversed targets sharply degrade response accuracy, showing that the gain depends on scientifically aligned sensitivity information rather than generic Jacobian regularization (Appendix F).

5 Related Work

Schrödinger bridges and continuous-time transport. Schrödinger bridges connect endpoint marginals while remaining close to a reference process [Léonard, 2014], with entropy-regularized optimal transport providing a closely related viewpoint [Cuturi, 2013]. Modern neural approaches include deep generative SBs [Wang et al., 2021], diffusion/IPF solvers [De Bortoli et al., 2021], conditional bridges [Shi et al., 2022], image-to-image bridges [Liu et al., 2023], closed-form Gaussian bridges [Bunne et al., 2023], DSBM/iterative Markovian fitting [Shi et al., 2023], and simulation-free score/flow matching [Tong et al., 2024]. Related diffusion, score-SDE, neural-SDE, flow-matching, and stochastic-interpolant models provide other continuous-time transport formulations [Ho et al., 2020, Song et al., 2021, Kidger et al., 2021, Lipman et al., 2023, Albergo et al., 2023]. These works primarily address how to construct, solve, or condition the transport; our question is how to constrain its sensitivity to changes in a scientific intervention.

Structured scientific transport. Generalized SB formulations inject state or control costs [Liu et al., 2022, 2024], while physics-aligned bridge methods enforce governing-equation or boundary consistency [Li et al., 2025]. More broadly, physics-informed networks and neural operators encode scientific structure through equations or parameter-to-solution maps [Raissi et al., 2019, Li et al., 2021]. These approaches constrain states, trajectories, costs, or solution operators. Tangent-SBM instead constrains the sensitivity of the learned stochastic transport with respect to an external intervention, which is useful when scientific information is available as a local sensitivity rather than a full target trajectory or terminal law.

Derivative and sensitivity supervision. Sobolev training, Jacobian matching, and derivative-informed neural operators show that derivative labels can contain information beyond function values [Czarnecki et al., 2017, Srinivas and Fleuret, 2018, O’Leary-Roseberry et al., 2024]. Derivative-informed operators supervise derivatives of deterministic parameter-to-solution maps, whereas we use derivative information to constrain pathwise and conditional-mean sensitivities of a learned stochastic transport. Related neural-SDE work develops scalable pathwise and adjoint sensitivity computation [Li et al., 2020]. Our contribution is not a new differentiation rule: we use tangent/JVP propagation to turn externally supplied scientific responses into supervision for the stochastic transport itself, and distinguish pathwise targets from conditional-mean sensitivity targets, for which ordinary derivative MSE is statistically inappropriate.

6 Discussion and Limitations

Sensitivity supervision is only as reliable as its target: shuffled and sign-reversed controls show that incorrect sensitivities can steer the model in the wrong direction. Moreover, first-order sensitivities are local; the PDEBench finite-perturbation and double-well Far-OOD results should therefore be interpreted as consistency tests rather than guarantees of indefinite extrapolation. The sensitivity-augmented objective falls outside standard unconstrained DSBM/IMF fixed-point theory; its path-space characterization remains future work.

7 Conclusion

TANGENT-SBM augments stochastic bridge learning with mechanistic intervention sensitivities. Across controlled systems and PDEBench, this substantially improves intervention-response accuracy while preserving endpoint fit, including under simulator-budget and finite-difference controls.

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Table 2: Complete Gaussian results, mean $\pm$ std over three seeds.

Split	Method	Endpoint RMSE	E_J	Finite resp.
Seen	DSBM	0.559 ± 0.001	1.000	0.345
	Conditional	0.245 ± 0.008	1.048 ± 0.007	0.357 ± 0.003
	TANGENT-SBM	0.244 ± 0.008	0.035 ± 0.004	0.017 ± 0.005
ID	DSBM	0.714 ± 0.002	1.000	0.321
	Conditional	0.440 ± 0.013	0.866 ± 0.003	0.275 ± 0.001
	TANGENT-SBM	0.249 ± 0.006	0.050 ± 0.005	0.016 ± 0.001
Near OOD	DSBM	0.511 ± 0.004	1.000	0.237
	Conditional	0.481 ± 0.036	1.247 ± 0.034	0.249 ± 0.004
	TANGENT-SBM	0.245 ± 0.007	0.129 ± 0.025	0.021 ± 0.004
Far OOD	DSBM	1.397 ± 0.006	1.000	0.512
	Conditional	0.397 ± 0.028	0.664 ± 0.008	0.329 ± 0.004
	TANGENT-SBM	0.260 ± 0.009	0.137 ± 0.026	0.085 ± 0.006

Table 3: Complete double-well results, mean $\pm$ std over three seeds.

Split	Method	Endpoint RMSE	E_J	Finite resp.
Seen	DSBM	0.392 ± 0.004	1.000	0.092
	Conditional	0.174 ± 0.007	0.459 ± 0.002	0.040 ± 0.002
	TANGENT-SBM	0.168 ± 0.003	0.271 ± 0.026	0.022 ± 0.003
ID	DSBM	0.336 ± 0.005	1.000	0.110
	Conditional	0.186 ± 0.006	0.341 ± 0.030	0.035 ± 0.005
	TANGENT-SBM	0.177 ± 0.002	0.250 ± 0.022	0.025 ± 0.003
Near OOD	DSBM	0.500 ± 0.002	1.000	0.042
	Conditional	0.168 ± 0.008	1.205 ± 0.224	0.045 ± 0.007
	TANGENT-SBM	0.166 ± 0.004	0.303 ± 0.005	0.013 ± 0.001
Far OOD	DSBM	0.542 ± 0.005	1.000	0.023
	Conditional	0.167 ± 0.007	2.549 ± 0.299	0.055 ± 0.006
	TANGENT-SBM	0.166 ± 0.006	0.984 ± 0.216	0.022 ± 0.007

Table 4: Complete PDEBench results, mean $\pm$ std over three seeds.

Split	Method	Field rel. L_2	E_J	Finite resp.
Seen	DSBM	0.715 ± 0.029	1.000	1.000
	Conditional	0.648 ± 0.033	0.828 ± 0.019	0.833 ± 0.016
	COND.+SPREADBUDGET-800	0.683 ± 0.026	0.861 ± 0.030	0.867 ± 0.024
	COND.+SPREADBUDGET-2720	0.699 ± 0.049	0.882 ± 0.082	0.892 ± 0.077
	TANGENT-SBM	0.624 ± 0.072	0.479 ± 0.048	0.493 ± 0.049
ID	DSBM	0.612 ± 0.034	1.000	1.000
	Conditional	0.605 ± 0.038	0.847 ± 0.054	0.843 ± 0.054
	COND.+SPREADBUDGET-800	0.646 ± 0.039	0.875 ± 0.033	0.872 ± 0.036
	COND.+SPREADBUDGET-2720	0.661 ± 0.063	0.855 ± 0.109	0.855 ± 0.111
	TANGENT-SBM	0.598 ± 0.069	0.508 ± 0.046	0.510 ± 0.046
Near OOD	DSBM	0.740 ± 0.036	1.000	1.000
	Conditional	0.668 ± 0.020	0.882 ± 0.022	0.878 ± 0.022
	COND.+SPREADBUDGET-800	0.699 ± 0.015	0.906 ± 0.025	0.904 ± 0.028
	COND.+SPREADBUDGET-2720	0.713 ± 0.034	0.945 ± 0.102	0.941 ± 0.101
	TANGENT-SBM	0.641 ± 0.063	0.490 ± 0.048	0.492 ± 0.045
Far OOD	DSBM	0.853 ± 0.032	1.000	1.000
	Conditional	0.729 ± 0.024	0.864 ± 0.065	0.867 ± 0.070
	COND.+SPREADBUDGET-800	0.755 ± 0.017	0.924 ± 0.042	0.922 ± 0.042
	COND.+SPREADBUDGET-2720	0.766 ± 0.028	0.896 ± 0.086	0.898 ± 0.087
	TANGENT-SBM	0.679 ± 0.066	0.547 ± 0.044	0.548 ± 0.045

B Simulator-Budget Controls

The PDEBench sensitivity-supervision set contains 108 settings at the endpoint-training anchors and 216 additional settings sampled continuously over the sensitivity-supervision region. Each directional sensitivity target requires two additional simulator evaluations, $F(X_0, a + \epsilon r)$ and $F(X_0, a - \epsilon r)$, giving

$$2(108 + 216) = 648$$

simulator calls beyond the original 270 endpoint-training pairs.

To test whether the improvement from TANGENT-SBM can be explained simply by these additional simulator evaluations, we construct endpoint-only controls with the same simulator-call budget.

COND.+LOCALBUDGET gives Conditional DSBM the exact $+\epsilon r$ and $-\epsilon r$ simulator outputs used to construct the sensitivity targets, but treats them as ordinary endpoint-training pairs. Thus LocalBudget and TANGENT-SBM use the same additional simulator evaluations at the same parameter settings; they differ in whether those evaluations are used as endpoint observations or combined into explicit directional sensitivity targets.

COND.+SPREADBUDGET instead allocates the same 648 additional simulator calls to ordinary endpoint examples sampled broadly over $[-1.3, 1.3]^3$, using fresh initial conditions from the same PDEBench initial-condition generator. This control tests whether the simulator budget would be more useful if it were spent on broader endpoint coverage rather than on local perturbation pairs. Evaluation for all variants uses the same fixed held-out split seeds.

Adding the 648 SpreadBudget examples enlarges the endpoint-training set from 270 to 918 examples. We therefore evaluate two continuation lengths. SPREADBUDGET-800 uses the same 800 continuation updates as the standard Conditional DSBM continuation. SPREADBUDGET-2720 instead approximately preserves the average number of optimization updates per training example by scaling the number of updates with the increase in dataset size:

$$800 \times \frac{918}{270} \approx 2720.$$

The two variants therefore test both equal update count and roughly equal optimization exposure per example.

Neither LocalBudget nor either SpreadBudget variant approaches the sensitivity accuracy obtained with explicit tangent supervision. The ordering of the two SpreadBudget training lengths varies across evaluation splits, so we report both rather than selecting one post hoc. We also do not interpret the fact

Table 5: **PDEBench simulator-budget controls.** All controls use the same additional simulator-call budget as sensitivity supervision. Results are mean $\pm$ standard deviation over three seeds. Lower is better.

Split	Control	Field rel. ℓ_2	E_J	Finite resp.
Seen	LocalBudget	0.670 ± 0.052	0.858 ± 0.087	0.864 ± 0.080
	SpreadBudget-800	0.683 ± 0.026	0.861 ± 0.030	0.867 ± 0.024
	SpreadBudget-2720	0.699 ± 0.049	0.882 ± 0.082	0.892 ± 0.077
ID	LocalBudget	0.636 ± 0.071	0.849 ± 0.094	0.846 ± 0.094
	SpreadBudget-800	0.646 ± 0.039	0.875 ± 0.033	0.872 ± 0.036
	SpreadBudget-2720	0.661 ± 0.063	0.855 ± 0.109	0.855 ± 0.111
Near OOD	LocalBudget	0.686 ± 0.040	0.908 ± 0.055	0.904 ± 0.055
	SpreadBudget-800	0.699 ± 0.015	0.906 ± 0.025	0.904 ± 0.028
	SpreadBudget-2720	0.713 ± 0.034	0.945 ± 0.102	0.941 ± 0.101
Far OOD	LocalBudget	0.738 ± 0.039	0.875 ± 0.081	0.877 ± 0.085
	SpreadBudget-800	0.755 ± 0.017	0.924 ± 0.042	0.922 ± 0.042
	SpreadBudget-2720	0.766 ± 0.028	0.896 ± 0.086	0.898 ± 0.087

that some endpoint-only controls perform slightly worse than plain Conditional DSBM as evidence for TANGENT-SBM. The relevant comparison is that neither using the exact perturbation simulations as endpoint data nor allocating the same simulator-call budget to broader endpoint coverage recovers the sensitivity gains obtained from explicit sensitivity supervision.

C Conditional-Mean Sensitivity Variance Ablation

Table 6: Double-well expected-response ablation, mean $\pm$ std over three seeds. Variance ratio is model conditional $\text{Var}(R_T)$ divided by simulator conditional $\text{Var}(R_T)$; one is ideal, but the two-rollout objective does not explicitly target this ratio.

Split	Objective	Mean-resp. RMSE	Variance ratio
Seen	Two-rollout	0.071 ± 0.014	0.698 ± 0.077
	Single-rollout MSE	0.097 ± 0.021	0.286 ± 0.043
ID	Two-rollout	0.066 ± 0.014	0.665 ± 0.102
	Single-rollout MSE	0.104 ± 0.025	0.312 ± 0.055
Near	Two-rollout	0.035 ± 0.002	0.669 ± 0.045
	Single-rollout MSE	0.036 ± 0.010	0.164 ± 0.005
Far	Two-rollout	0.085 ± 0.021	1.012 ± 0.296
	Single-rollout MSE	0.104 ± 0.010	1.156 ± 0.183

D Implementation and Compute

All reported numbers use model seeds 32, 42, and 52. The sensitivity weights are selected by the sweeps reported in the main text and then fixed for all three-seed runs. For reproducibility, Gaussian and double-well variants share the conditional checkpoint after IMF 3 and are continued through IMF 7; PDEBench variants share the conditional checkpoint after IMF 1 and are continued through IMF 2.

Evaluation split supports. For the double-well benchmark, endpoint training uses $u \in \{-1, 0, 1\}$, ID uses $u \sim \mathcal{U}[-1, 1]$, Near OOD uses $|u| \sim \mathcal{U}[1.10, 1.30]$, and Far OOD uses $|u| \sim \mathcal{U}[1.40, 1.70]$; continuous response-supervision uses $u \sim \mathcal{U}[-1.30, 1.30]$. For PDEBench, ID samples $a \in [-1, 1]^3$, Near OOD samples the shell $1.10 \leq \|a\|_\infty \leq 1.30$, and Far OOD samples $1.40 \leq \|a\|_\infty \leq 1.70$; continuous response-supervision uses $a \in [-1.30, 1.30]^3$.

Gaussian and double-well learned drifts use width-128, depth-3 SiLU MLPs, 30 Euler–Maruyama steps, reference diffusion 0.50, 1200 inner updates, bridge batch size 512, and learning rate 10^{-4} ; sensitivity updates use batch size 256. The double-well simulator uses 200 Euler–Maruyama steps over $T = 2$, diffusion 0.45, and tilt scale 0.30.

PDEBench uses the official 128×128 diffusion–reaction simulator on $[-1, 1]^2$ with $T = 5$, 101 recorded time points, and default $D_u^0 = 10^{-3}$, $D_v^0 = 5 \times 10^{-3}$, $k^0 = 5 \times 10^{-3}$. The learned drift is a U-Net with base width 32, 20 learned-SDE steps, reference diffusion 0.15, batch size 8, learning rate 2×10^{-4} , and 800 standard updates. Tangent training uses response batch size 4, and applies response matching at every forward update. The sensitivity-weight sweeps are $\{0.05, 0.10, 0.25, 0.50\}$ for Gaussian, $\{0.10, 0.25, 0.50, 1.0\}$ for double well, and $\{0, 0.1, 0.5, 1, 2, 5\}$ for PDEBench, with selected values 0.5, 0.25, and 2.0, respectively. Sensitivity weights were selected using a held-out validation set disjoint from all Seen/ID/Near/Far evaluation samples, based on the endpoint–response tradeoff, and were then fixed for all three-seed test results.

Measured PDEBench continuation wall times on the same setup average 84.7 s for SpreadBudget-800, 134.1 s for SpreadBudget-2720, and 515.1 s for Tangent-SBM. Thus Tangent-SBM is about $6.1 \times$ slower than the 800-update spread endpoint control in this implementation. These times include evaluation and are not hardware-independent complexity estimates. Simulator-call budget is matched by the endpoint controls, while wall-clock compute is not. We do not extend endpoint-only training to wall-clock parity because additional unconstrained IMF continuation already degrades PDEBench endpoint behavior in this pipeline.

Code and experiment-generation scripts will be released with the public version.

E Gaussian Directionality

For the Gaussian system, the reverse conditional is analytic, allowing an independently valid reverse-response target. On ID, endpoint RMSE is 0.440 ± 0.013 for conditional DSBM, 0.238 ± 0.008 for forward-only, 0.427 ± 0.010 for backward-only, and 0.219 ± 0.010 for bidirectional supervision. The corresponding forward response errors are 0.866, 0.052, 0.869, and 0.066. Forward and reverse constraints therefore encode distinct response information.

F Additional Ablations

Response amount. For Gaussian, all runs use $\lambda_{\text{sens}} = 0.5$. ID E_J is 0.050 ± 0.005 , 0.050 ± 0.005 , and 0.050 ± 0.005 with 25%, 50%, and 100% of continuous response-supervision; Far-OOD error is 0.142 ± 0.029 , 0.139 ± 0.027 , and 0.137 ± 0.026 . For double well ID, the corresponding errors are approximately 0.249, 0.250, and 0.250. For PDEBench ID they are approximately 0.516, 0.509, and 0.508.

Target correctness. On Gaussian ID, E_J is 0.050 ± 0.005 with correct targets, 0.858 ± 0.007 when shuffled, and 1.817 ± 0.008 when sign-reversed. On PDEBench, shuffled targets return response error to approximately one across splits, while sign reversal gives errors of roughly 1.55–1.65 and substantially degrades field prediction. Double well shows the same qualitative dependence on correct target alignment.

Exact Gaussian parameters. The benchmark uses

$$A = \begin{bmatrix} 0.80 & -0.30 \\ 0.25 & 0.65 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 0.70 \\ -0.40 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0.25 \\ 0.15 \end{bmatrix}, \quad B_3 = \begin{bmatrix} 0.55 \\ -0.45 \end{bmatrix},$$

with

$$\Sigma_0 = \begin{bmatrix} 1.00 & 0.35 \\ 0.35 & 0.80 \end{bmatrix}, \quad \Sigma_\epsilon = \begin{bmatrix} 0.25 & 0.08 \\ 0.08 & 0.18 \end{bmatrix}.$$

G Finite-Difference Response Matching

To test whether the sensitivity gains require explicit tangent propagation, we construct a numerical finite-difference response control on PDEBench. For a response-supervision tuple (X_0, a, v) , we replace the learned-model directional tangent $J_\theta v$ by

$$\hat{J}_{\text{FD}} v = \frac{Y_\theta(a + \delta v, \omega) - Y_\theta(a - \delta v, \omega)}{2\delta}, \quad (18)$$

where the positive and negative perturbation rollouts use identical Brownian increments. This common-noise coupling isolates the effect of changing the intervention rather than introducing additional stochastic variation.

Table 7: PDEBench finite-difference response-matching control under a matched stochastic-rollout budget. Values are mean $\pm$ standard deviation over three seeds. The final column averages the four split means. Lower is better.

Method	Seen	ID	Near OOD	Far OOD	Avg.
FD, $\delta = 0.01$	0.505 ± 0.011	0.533 ± 0.012	0.517 ± 0.002	0.575 ± 0.015	0.533
FD, $\delta = 0.03$	0.494 ± 0.024	0.520 ± 0.021	0.502 ± 0.019	0.560 ± 0.007	0.519
FD, $\delta = 0.10$	0.487 ± 0.022	0.517 ± 0.017	0.497 ± 0.021	0.553 ± 0.010	0.514
FD, $\delta = 0.25$	0.494 ± 0.034	0.525 ± 0.031	0.499 ± 0.035	0.559 ± 0.024	0.519
TANGENT-SBM	0.479 ± 0.048	0.508 ± 0.046	0.490 ± 0.048	0.547 ± 0.044	0.506

The finite-difference estimate replaces $J_{\theta}v$ in the same response objective used by TANGENT-SBM. The bridge checkpoint, continuation schedule, sensitivity targets, sensitivity weight, and response-update frequency are otherwise unchanged. A finite-difference response requires two perturbed learned-SDE trajectories per response example. We therefore use response batch size 2 for the finite-difference control and batch size 4 for TANGENT-SBM, matching the number of stochastic state rollouts per response update. Evaluation is identical across methods and uses the same directional-sensitivity metric defined in Eq. 17; finite differences are used only as the training mechanism.

Centered differences introduce a perturbation-scale parameter δ , so we evaluate $\delta \in \{0.01, 0.03, 0.10, 0.25\}$ rather than selecting a single favorable value. Table 7 reports the resulting sensitivity errors.

Finite-difference response matching is itself effective: all tested values of δ substantially improve over Conditional DSBM, confirming that the primary gain comes from explicitly constraining intervention response rather than conditioning on the physical parameter alone. With a well-chosen perturbation scale, finite differences attain sensitivity accuracy comparable to TANGENT-SBM: the strongest setting, $\delta = 0.10$, obtains average $E_J = 0.514$, compared with 0.506 for TANGENT-SBM. Given the three-seed variability, we do not interpret this difference as a statistically meaningful accuracy advantage. The practical distinction is that tangent propagation directly represents the infinitesimal response and does not require selecting a finite-difference scale.